

Day -3 : Introduction to IP Addressing Theory

1. Key Differences Between IPv4 and IPv6 :

Feature	IPv4	IPv6
Address Size	32-bit address length	128-bit address length
Address Format	Dotted-decimal notation (e.g., 192.168.1.1)	Hexadecimal notation separated by colons (e.g., 2001:db8::1)
Total Address Space	$\sim 4.3 \times 10^9$ addresses (~ 4.3 billion)	$\sim 3.4 \times 10^{38}$ addresses (virtually unlimited)
Configuration	Configured manually or automatically via DHCP	Supports Auto-configuration (SLAAC) without needing a DHCP server

2. Static IPv4 Configuration (192.168.10.0/24) :

Assuming the router (Default Gateway) uses 192.168.10.1:

- **Laptop**
 - **IP Address:** 192.168.10.10
 - **Subnet Mask:** 255.255.255.0
 - **Default Gateway:** 192.168.10.1
- **Phone**
 - **IP Address:** 192.168.10.11
 - **Subnet Mask:** 255.255.255.0
 - **Default Gateway:** 192.168.10.1
- **Smart TV**
 - **IP Address:** 192.168.10.12
 - **Subnet Mask:** 255.255.255.0
 - **Default Gateway:** 192.168.10.1
- **Printer**
 - **IP Address:** 192.168.10.13
 - **Subnet Mask:** 255.255.255.0
 - **Default Gateway:** 192.168.10.1

3. Subnetting 192.168.1.0/27 :

- **Number of Valid Subnets:** $2^{(27-24)} = 2^3 = 8$ subnets
- **Usable Hosts Per Subnet:** $2^{(32-27)} - 2 = 2^5 - 2 = 30$ hosts per subnet

First Three Subnets:

- **Subnet 1:**
 - **Network Address:** 192.168.1.0
 - **Usable Host Range:** 192.168.1.1 - 192.168.1.30
 - **Broadcast Address:** 192.168.1.31
- **Subnet 2:**
 - **Network Address:** 192.168.1.32
 - **Usable Host Range:** 192.168.1.33 - 192.168.1.62
 - **Broadcast Address:** 192.168.1.63
- **Subnet 3:**
 - **Network Address:** 192.168.1.64
 - **Usable Host Range:** 192.168.1.65 - 192.168.1.94
 - **Broadcast Address:** 192.168.1.95

4. Subnetting 10.0.5.17 with Mask 255.255.255.0 (Simplified) :

Think of the subnet mask 255.255.255.0 as a filter: the first three numbers (10.0.5) belong to the neighborhood, and the last number identifies the specific house.

- **Network Address (Neighborhood ID):** 10.0.5.0 (This names the entire local network).
- **Broadcast Address (Shout-out Address):** 10.0.5.255 (Used to send a message to every device on this network at once).
- **Usable Host Addresses (Assignables):** 10.0.5.1 through 10.0.5.254 (These are the actual IP addresses you can give to laptops, phones, or printers).

5. Verification Statement :

When you plug these details into an online subnet calculator, the results line up completely:

- **Verification:** The manual math matches the online calculator's results exactly, confirming 10.0.5.0 as the network address, 10.0.5.255 as the broadcast address, and 10.0.5.1 to 10.0.5.254 as the usable device range.